

Harpreet Singh

DATA ENGINEERING | PYTHON PIPELINES | SCIENTIFIC AND SENSOR DATA

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Professional Summary

Research and embedded systems engineer focused on Python data pipelines, time-series acquisition, dataset preparation, and reproducible machine-learning workflows. Builds workflows that capture EEG and sensor streams, synchronize event labels, transform raw recordings into analysis-ready datasets, and export model artifacts. Combines practical hardware integration with data preprocessing, session metadata, diagnostic logging, and cybersecurity research.

Technical Skills

Programming and analysis: Python, NumPy, Pandas, Bash, C, scikit-learn, PyTorch, TensorFlow/Keras.

Data pipelines: Stream ingestion, batch preprocessing, extract-transform-export workflows, time-series alignment, event labels, feature engineering, CSV, JSON, XDF.

Reliability and reproducibility: Preprocessing manifests, session metadata, diagnostic logs, stream discovery, configuration-driven workflows, held-session-out evaluation, Git.

Acquisition and integration: Lab Streaming Layer (LSL), LabRecorder, CGX EEG, UDP, HTTP, Raspberry Pi, Linux, SPI/GPIO, Zeek, Wireshark.

Selected Data Engineering Projects

CGX Dataset and Game | EEG acquisition and dataset pipeline

github.com/baazSingh13/CGX_dataset_game

- Built a Python workflow connecting CGX EEG streams and five-state visual-stimulus markers to LabRecorder for synchronized XDF recording.
- Transformed XDF sessions into labelled CSV datasets; recorded selected streams, channel indices, sample rate, reference, filters, and exported sample counts in preprocessing manifests.
- Connected dataset preparation to classifier training, held-session-out metrics, model metadata, and quantized model exports for downstream software and hardware workflows.
- Integrated background startup tasks and status polling to expose acquisition failures; connected live classifier output to a browser game through a defined UDP packet contract.

Sheeg | Recording orchestration and stream diagnostics

github.com/baazSingh13/Sheeg

- Developed launch and recording workflows around CGX EEG, behavioural tasks, LSL event markers, and LabRecorder, with session directories and diagnostic output.
- Documented a marker schema and added separate EEG/marker discovery diagnostics to investigate missing event streams and session-alignment failures.

EEG and fNIRS Platform | Sensor ingestion and logging

- Developed Raspberry Pi acquisition software for 8-channel EEG and 8-channel analog acquisition, integrating ADS1299, ADS8688, and DAC8565 devices.
- Coordinated shared SPI access using threading locks, GPIO/MUX sequencing, live visualization, and CSV logging to support concurrent acquisition and offline analysis.

Professional Experience

Research Assistant | University of Lethbridge | Apr 2025 - Present

- Develop research data workflows spanning EEG/fNIRS acquisition, behavioural event synchronization, labelled dataset creation, Python analysis, and model export.
- Integrate hardware acquisition with CGX/Sheeg recording tools, session metadata, preprocessing scripts, and live classification interfaces.
- Prepare temporal neural data for FPGA-oriented experiments, mapping 12 time bins by 196 features into fixed-point inputs with software reference checks.

Independent Embedded Systems Developer | Self-Employed | Jun 2024 - Mar 2025

- Designed a custom STM32F405RGT6 MicroPython development board and integrated sensor/actuator interfaces, firmware, and hardware debugging workflows.

Research Assistant | University of Regina | Jan 2021 - Apr 2024

- Prepared network-traffic features and Python machine-learning workflows for DoS/DDoS intrusion detection using Zeek, Wireshark, Bayesian Networks, and Markov Networks.
- Conducted preprocessing, feature engineering, model tuning, and experimental evaluation for uncertain-reasoning intrusion detection research published at SECRIPT 2024.
- Built additional medical-diagnosis and Punjabi NLP classification prototypes, including work with a corpus of more than 100,000 news articles.

Senior Software Engineer | Planetcast Media Services Ltd. | Dec 2018 - Nov 2020

- Led a 15-member team delivering embedded hardware and firmware for broadcast/media systems, including a network-controlled 10-channel video switcher and Ethernet temperature-acquisition platform.
- Developed Ethernet and HTTP/LWIP interfaces for remote monitoring, alarms, and device control, connecting physical measurements to networked applications.

Embedded Software Engineer / R&D Engineer | Spark Eighteen; Exicom | Feb 2016 - Dec 2018

- Developed STM32/AVR systems for LiFi, telecom power monitoring, interface cards, and automated test equipment; integrated CAN, SPI, I2C, and Ethernet interfaces.
- Performed board bring-up, firmware validation, and GUI-based test automation for industrial monitoring and control systems.

Education

PhD in Computational Behavioural Neuroscience | University of Lethbridge | In progress

M.Sc. Computer Science | University of Regina | 2024

Bachelor's equivalent in Electronics & Telecommunication Engineering | AMIETE/IETE | 2012

Research Publications

Singh, H., et al. "An Uncertain Reasoning-Based Intrusion Detection System for DoS/DDoS Detection." SECRIPT 2024, pp. 771-776.

Singh, H. A robust intrusion detection system utilizing uncertain reasoning techniques in artificial intelligence. M.Sc. thesis, University of Regina, 2024.